

Exponenciální model $T \sim \text{Exp}(\lambda); \lambda > 0$

$$T \sim f_T(t) = \lambda e^{-\lambda t} \quad ; \quad t > 0$$

$$R_T(t) = e^{-\lambda t} \quad ; \quad \text{MTTF} = \left(\frac{1}{\lambda}\right) !$$

$$R_T(1/\lambda) = e^{-1} \approx 0,3679$$

cFR: $\lambda_T(t) = \lambda$ konst. cFR IFR i DFR

cFRA: $L_T(t) = \int_0^t \lambda dt' = \lambda \cdot t \Rightarrow L_T/t$ konst.

NBU i NWU: $R(x|t) = \frac{P(T > t+x)}{P(T > t)} = e^{-\lambda x} = R_T(x)$

nezávisí na minulosti

NBUE i NWUE: $\overset{\curvearrowright}{MLR}(t) = \int_0^\infty R(x|t) dx = \int_0^\infty R_T(x) dx = \text{MTTF}$

bez paměti, věk výrobku nemá vliv!

odhady $\hat{\lambda}$: $\hat{\lambda}_{ML} = \left(\frac{1}{n} \overset{\text{TTT total time in test}}{\sum_{j=1}^n T_j} \right)^{-1} = \overline{T}_n^{-1} = \frac{1}{\overline{T}(T_n)} \cdot n$

Bayes: $\pi(\lambda) = \text{gamma}(a, b)$ Conjugated fam.

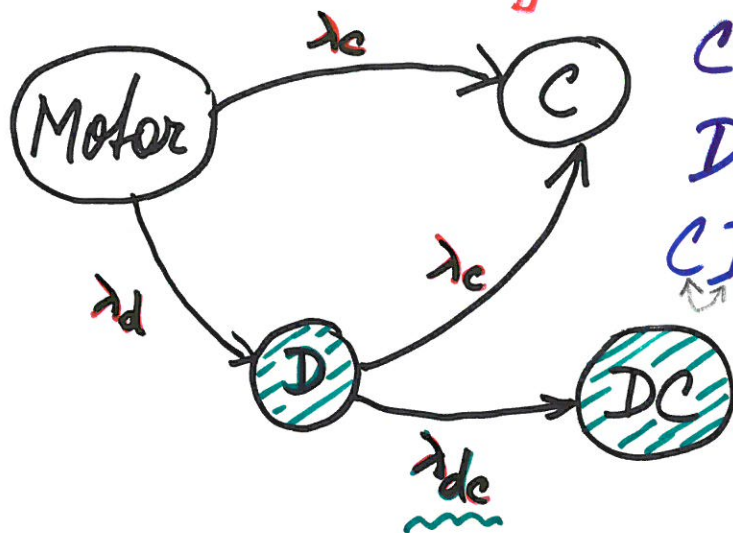
$$\hat{\lambda}_B \stackrel{L_2}{=} \frac{n+a}{\sum_{j=1}^n t_j + b}$$

není invariantní
na transf.

$$\hat{R}_B(t) \neq R_T(\hat{\lambda}_B) = e^{-\hat{\lambda}_B \cdot t}$$

Př: PHASE-TYPE DISTRIBUTION

State transition diagram



C kritická porucha
D degradační fáze/mod
CD degrad. kritická porucha

$T_c \dots \lambda_c$; $T_d \dots \lambda_d$; $T_{dc} \dots \lambda_{dc}$ nezávisle

$$T_s = \min \{ T_c, T_d + T_{dc} \}$$

$$R_s(t) = P(\min \{ T_c, T_d + T_{dc} \} > t) = P(T_c > t, T_d + T_{dc} > t)$$

$$\stackrel{\text{id}}{=} P(T_c > t) \cdot P(T_d + T_{dc} > t) = \text{dos. + konvoluce}$$

$$= e^{-\lambda_c t} \cdot \frac{1}{\lambda_{dc} - \lambda_d} (\lambda_{dc} e^{-\lambda_d t} - \lambda_d e^{-\lambda_{dc} t})$$

⇓

$$\lambda_s(t) = -R'_s/R_s = \lambda_c + \frac{\lambda_d \lambda_{dc} (e^{-\lambda_d t} - e^{-\lambda_{dc} t})}{\lambda_{dc} e^{-\lambda_d t} - \lambda_d e^{-\lambda_{dc} t}}$$

$$MTTF_s = \int_0^\infty R_s(t) dt = \frac{\lambda_c + \lambda_d + \lambda_{dc}}{(\lambda_c + \lambda_d)(\lambda_c + \lambda_{dc})}$$

R_T^v : MIXTURE MODELS

Dva závody - výroba - výrobek - s rozdílnou technologií
(různá kvalita)

1. závod $T_1 \dots R_{T_1} = R_1$ s CFR λ_1
2. závod $T_2 \dots R_2$ s CFR λ_2 } $\lambda_1 \neq \lambda_2$

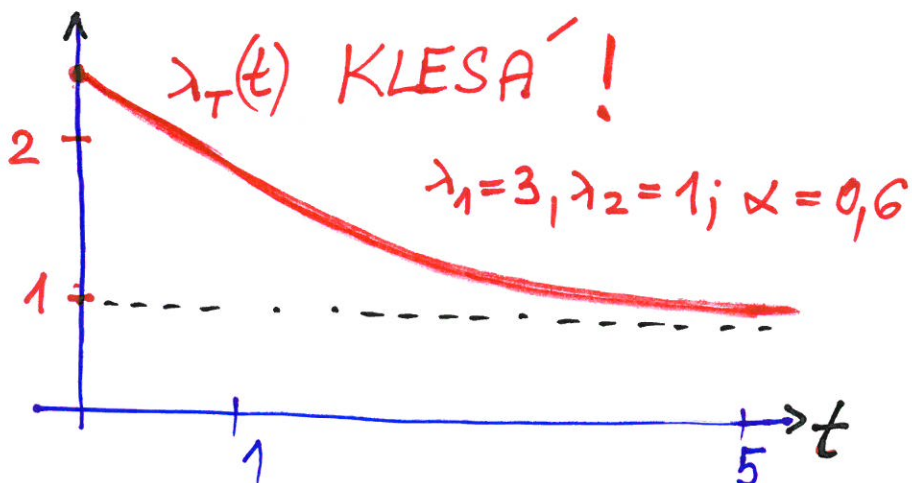
Výrobky smíchány před prodejem $\alpha : (1-\alpha)$

Náhodně zakoupíme 1 výrobek $\Rightarrow R_T$, MTTF?

$$R_T(t) = \alpha R_1(t) + (1-\alpha) R_2(t) = ; t \in \mathbb{R}$$
$$= \alpha e^{-\lambda_1 t} + (1-\alpha) e^{-\lambda_2 t}$$

$$\Downarrow \text{MTTF} = \int_0^{\infty} R_T dt = \frac{\alpha}{\lambda_1} + (1-\alpha)/\lambda_2$$

$$a ! \lambda_T(t) = -R_T' / R_T = \frac{\alpha \lambda_1 e^{-\lambda_1 t} + (1-\alpha) \lambda_2 e^{-\lambda_2 t}}{\alpha e^{-\lambda_1 t} + (1-\alpha) e^{-\lambda_2 t}}$$



$$\lim_{t \rightarrow \infty} \lambda_T(t) = \min(\lambda_1, \lambda_2)$$

Zobecnění: $R_T = \sum_j \alpha_j \cdot R_j \Rightarrow \lambda_T = \frac{\sum_j \alpha_j \lambda_j e^{-\lambda_j t}}{\sum_j \alpha_j e^{-\lambda_j t}}$

SMIRNOVova TRASFORMace :

$T \sim R_T$. Def. $\tilde{T} = \Lambda_T(T) = -\ln(1 - F_T(T))$.

Pak $\tilde{T} \sim \text{Exp}(1) \wedge \lambda_{\tilde{T}}(t) = 1$ (cFR systém).

Navíc $T = t \Leftrightarrow \tilde{T} = \Lambda_T(t), \forall t \geq 0$.

Tzn. původní systém stárnoucí/mládnoucí s poruchou v čase t_0 je ekvival. se systémem $\tilde{T}$ nestárnoucím s poruchou v čase $\Lambda_T(t_0) \triangleq$ EXPOnciální REZERVA systému T do mom. t .

$$\text{Dk: } P(F_T(T) \leq t) = P(T \leq F_T^{\leftarrow}(t)) = F_T(F_T^{\leftarrow}(t)) = t \in (0,1)$$

$$\Downarrow R_{\tilde{T}}(t) = P(\tilde{T} > t) = P(-\ln(1 - \underbrace{F_T(T)}_{U \sim U(0,1)}) > t) = \\ = P(U > 1 - e^{-t}) = 1 - F_U(1 - e^{-t}) = e^{-t}, t \geq 0$$

Pokud $\lambda_T > 0 \Rightarrow \Lambda_T(t) = \int_0^t \lambda_T(t') dt'$ ostře roste

Aedy $T = t_0$ jednoznačně určuje $\Lambda_T(t_0)$ + spojitě

Pozn: Λ_T v praxi neznáme!

bud' máme fyzik. zákon nebo ho lze odhadnout z dat

PARETO MODEL

$$T \sim R_T(t) = \frac{\beta^k}{(\beta+t)^k} = \left(1 + \frac{t}{\beta}\right)^{-k}; \quad \begin{matrix} \beta > 0, \\ k > 0, \\ t > 0. \end{matrix}$$

$$\Downarrow f_T(t) = -R_T' = \frac{k\beta^k}{(\beta+t)^{k+1}} = \frac{k}{\beta} \left(1 + \frac{t}{\beta}\right)^{-k}$$

$$\Downarrow \lambda_T(t) = f_T/R_T = \frac{k}{\beta+t}$$

DFR monotonně

$$\lim_{t \rightarrow \infty} \lambda_T = 0$$

$$MTTF_T = \int_0^{\infty} R_T = \frac{\beta}{k-1}$$

! pouze pro $k > 1$ ($\exists$)

Obr.

Př! Výrobek... $T \sim \text{Exp}(\lambda)$; cFR $\lambda_T = \lambda$. [T/λ]

Nestabilní výroba $\Rightarrow \lambda \sim \text{gamma}(k, \beta)$ [π(λ)]

Bayes

$$\Downarrow R_T(t) = \int_t^{+\infty} \int_0^{+\infty} f_{T/\lambda} \cdot \pi(\lambda) d\lambda du^{t'} =$$

$$= \int_0^{\infty} \pi(\lambda) \cdot R_{T/\lambda} d\lambda = \int_0^{\infty} e^{-\lambda t} \frac{\beta^k}{\Gamma(k)} \lambda^{k-1} e^{-\beta\lambda} d\lambda =$$

$$= \frac{\beta^k}{\Gamma(k)} \int_0^{\infty} \lambda^{k-1} e^{-\lambda(t+\beta)} d\lambda = \frac{\beta^k}{(t+\beta)^k} \quad \text{[PARETO]}$$

Nestabilní výroba $\Rightarrow R_T$ polynomiálně klesá!
= Bayes

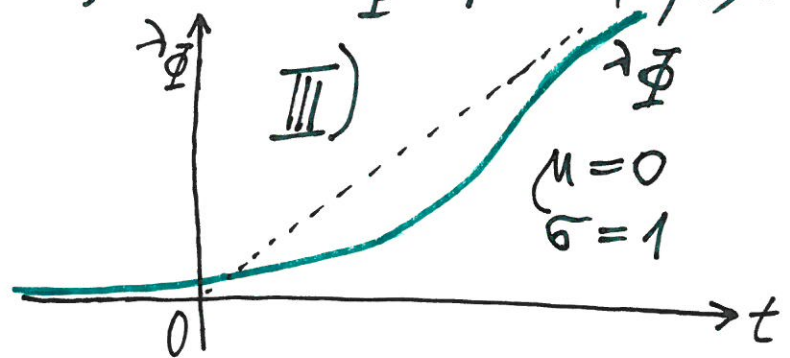
NORMAL MODEL ~ TS

$$T \sim R_T(t) = 1 - \Phi\left(\frac{t-\mu}{\sigma}\right) ; t \in \mathbb{R}!$$

$$\lambda_T(t) = -R_T'/R_T = \frac{1}{\sigma} \varphi\left(\frac{t-\mu}{\sigma}\right) / \left(1 - \Phi\left(\frac{t-\mu}{\sigma}\right)\right) = \\ = \frac{1}{\sigma} \lambda_{\Phi}\left(\frac{t-\mu}{\sigma}\right), \text{ kde } \lambda_{\Phi} \text{ odpov. } N(0,1).$$

$$\lim_{t \rightarrow \infty} \lambda_{\Phi}(t)/t = 1$$

$$\text{MTTF} = \mu, \text{DT} = \sigma^2.$$



Užití: doba opotřebení pneumatik $\perp P_{\text{ř.}}$

Výhoda: $T_S = \sum_{j=1}^L T_j$, T_j iid $\mathcal{L}_2$ s lib. rozdělením.

Pak dle CLT: $T_S \sim AN$ nebo REPRODUKCE.

TN: USEKNUTÉ (truncated) N zleva at $t=0$:

$$R_T^{\text{TR}}(t) = P(T > t | T > 0) = \frac{1 - \Phi\left(\frac{t-\mu}{\sigma}\right)}{1 - \Phi(\mu/\sigma)} ; t \geq 0.$$

$$\lambda_T^{\text{TR}}(t) = -R_T'/R_T = \frac{1}{\sigma} \lambda_{\Phi}\left(\frac{t-\mu}{\sigma}\right) ! \text{ pro } t \geq 0 !$$

Odhady $\hat{\mu}, \hat{\sigma}^2$ standard nebo $\propto TN(\mu, \sigma^2)$!

Gamma MODEL ~ TS

$$T \sim R_T(t) = \int_t^\infty \underbrace{\frac{\lambda}{\Gamma(k)} (\lambda t')^{k-1} e^{-\lambda t'}}_{\text{gamma}(k, \lambda)} dt' ; \quad \begin{matrix} \lambda > 0, \\ k \in \mathbb{N}(\mathbb{R}), \\ t > 0. \end{matrix}$$

$$ET = \text{MTTF} = k/\lambda$$

$$DT = k/\lambda^2$$

$$\lambda_T(t) = \frac{t^{k-1} e^{-\lambda t}}{\int_t^\infty t'^{k-1} e^{-\lambda t'} dt'}$$

Obt.

$$\lim_{t \rightarrow \infty} \lambda_T(t) = \lambda ! \quad \lim_{t \rightarrow 0+} \lambda_T(t) = \begin{cases} \infty & 0 < k < 1, \\ 0 & k > 1, \\ \lambda & k = 1. \end{cases}$$

$k=1$ Exp(λ) ; $k \in \mathbb{N}$ ERLANGOVO rozdělení

$\lambda \dots$ intenzita šoku ; $\lambda > 0$. [HPP]

$k \dots$ míra rezistence proti šokům ; $k > 0$.

$$k \in \mathbb{N} \Rightarrow \lambda_T(t) = \frac{\lambda (\lambda t)^{k-1} e^{-\lambda t} / \Gamma(k)}{\sum_{l=0}^{k-1} (\lambda t)^l e^{-\lambda t} / l!} \quad [\text{Polynom}^{-1}]$$

Reprodukce : $T_j \sim \text{Gamma}(k_j, \lambda) \Rightarrow$

$$T_s = \sum_j T_j \sim \text{Gamma}\left(\sum_j k_j, \lambda\right)$$

Užití : $0 \mid \begin{matrix} \times & \times & \times & \dots \\ T_1 & T_2 & T_3 & \dots \\ (k_1) & (k_2) & (k_3) & \dots \end{matrix}$ opravy
(plán opt. oprav)

Odhady $\hat{\lambda}, \hat{k}$: MM, MLE, ...

Log Gamma MODEL

$$T \sim LG(k, \lambda) \Leftrightarrow U = -\log T \sim G(k, \lambda)$$

Pozor: $T = e^{-U} \in (0, 1)$! TRANSFORM.

Alt: $T \sim G(k, \lambda) \Rightarrow Y = e^{-T} \sim LG(k, \lambda)$
 $Y \in (0, 1)$

(MEU) $f_Y(y) = \frac{\lambda^k}{\Gamma(k)} \cdot (-\ln y)^{k-1} \cdot y^{\lambda-1}$
"MTTF_Y" = $\left(\frac{\lambda}{\lambda+1}\right)^k$; DT existsuje...

Invert. gamma MODEL

$$T \sim IG(k, \lambda) \Leftrightarrow 1/T \sim G(k, \lambda); \begin{matrix} \lambda > 0 \\ k > 0 \end{matrix}$$

$$f_T(t) = \frac{\lambda^k}{\Gamma(k)} t^{-k-1} e^{-\lambda/t} ; \underline{t > 0}$$

Pozor: $ET = MTTF = \frac{\lambda}{k-1}$ if $k > 1$!
($\exists$)

MEU! $DT = \frac{\lambda^2}{(k-1)^2(k-2)}$ if $k > 2$!